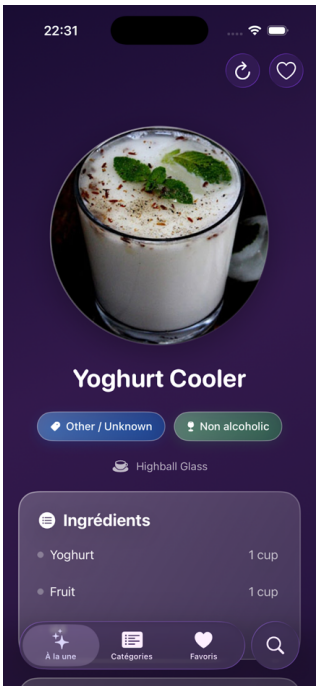
How to make the greatest cocktail app with baby steps

Part I : Discovering Android Studio and creating a first user interface

1. Initialize the project

- Create a new Android project in Android Studio. Use Empty Compose Activity and name the application TheGreatestCocktailApp. For the package name, use the following format: fr.isen.[name].thegreatestcocktailapp
- Configure the API level so that it includes at least 95% of users.
- Test the application on an emulator or your Android phone in developer mode.
- Familiarize yourself with the different components offered by jetpack compose in Android Studio (Text, Image, Button, Row, Column, etc.).
- Familiarize yourself with the life cycle of an activity inside the MainActivity. (onCreate, onRestart, onResume, onStop, onPause, onDestroy)
- Initialize the git project and create a repository on Github (Public) that will be linked to your project. Your Github project url will be required for the project assessment.

2. Design your first interface



- Add an image to the project
- Create the **DetailCocktailScreen.kt,** create a new Composable fun named DetailCocktailScreen inside**,** remove the Composable fun “Greeting” and plug the new screen on your MainActivity
- In the DetailCocktailScreen function, start adding the image, the title, the categories, the type of glass, the card of ingredients, and the card of recipe.
- Add a TopAppBar and an IconButton to add the cocktail to the favorite section
- When clicking on the favorite button, display a toast

3. Do some lists

- Create a new kotlin file named **CategoriesScreen.kt** with the Composable fun inside
- This fun needs to display a list of categories (Beer, cocktail, cocoa, coffee, shot, soft drink, etc...)
- When clicking on a category, start a new activity to display a list of drinks
- When clicking on a drink, start a new activity to display the detail cocktail screen (pay attention, you already have a DetailCocktailScreen).
- In the MainActivity, add a NavHost with 3 items: random, list, favorites

4. Time to take IPA

- Add RetroFit2 for the network request, Gson for your parsing, and Coil for the online image to your graddle.

For now, we will use thecocktaildb api : https://www.thecocktaildb.com/api/json/v1/1/
The endpoints are :

Random : `random.php`
List of categories : `list.php?c=list`
List of drinks for a category : `filter.php?c=YOUR_CATEGORY`
Detail of a drink : `lookup.php?i=YOUR_DRINK_IDENTIFIER`

- Create the Serializable objects to parse the response of all this api, based on the json.

- Create an interface named ApiService for listing all the api method above.
- Now just create an object for the network manager, that uses ApiService.
- Let’s test that by calling it in the function called each time the start activity is shown. Display at the DetailCocktailScreen with real data from the random api.
- Do the same thing for the list.

5. Let's add some love

- Replace the toast with the storage of the cocktail actually shown. (be aware the detail screen is displayed on 2 different activities). For the storage, let’s use the SharedPreference
- On the FavoriteScreen, display the list of favorite drinks
- When clicking on an element, show the DetailCocktailScreen.
- On the screen, if the user clicks on the favorite button, add or remove the cocktail to the SharedPreference.

6. Still have time ?

The api offers you some endpoints to upgrade your application, like a search by name or ingredient: https://www.thecocktaildb.com/api.php

7. Deployment

Generate your APK and don’t forget to push all your modifications on github.

Somes Helping links

Retrofit: http://medium.com/@anna972606/retrofit-in-android-15fa724a8fa6
Coil: https://developer.android.com/develop/ui/compose/graphics/images/loading

Submitted before March 8th at 11:59 PM